Problem

This project is an attempt to address the problems accompanying the physical and digital Blackjack environment. Many people are interested in casinos but don't have easy access to them. While there are digital casinos and places to play games like Blackjack, they are often too closely tied with real gambling platforms or have too many ads that may be uncomfortable or annoying. The goal is to create a Blackjack game that simulates a realistic, sleek game that delivers the highlights of a real Blackjack game without the negatives.

Background and Motivation

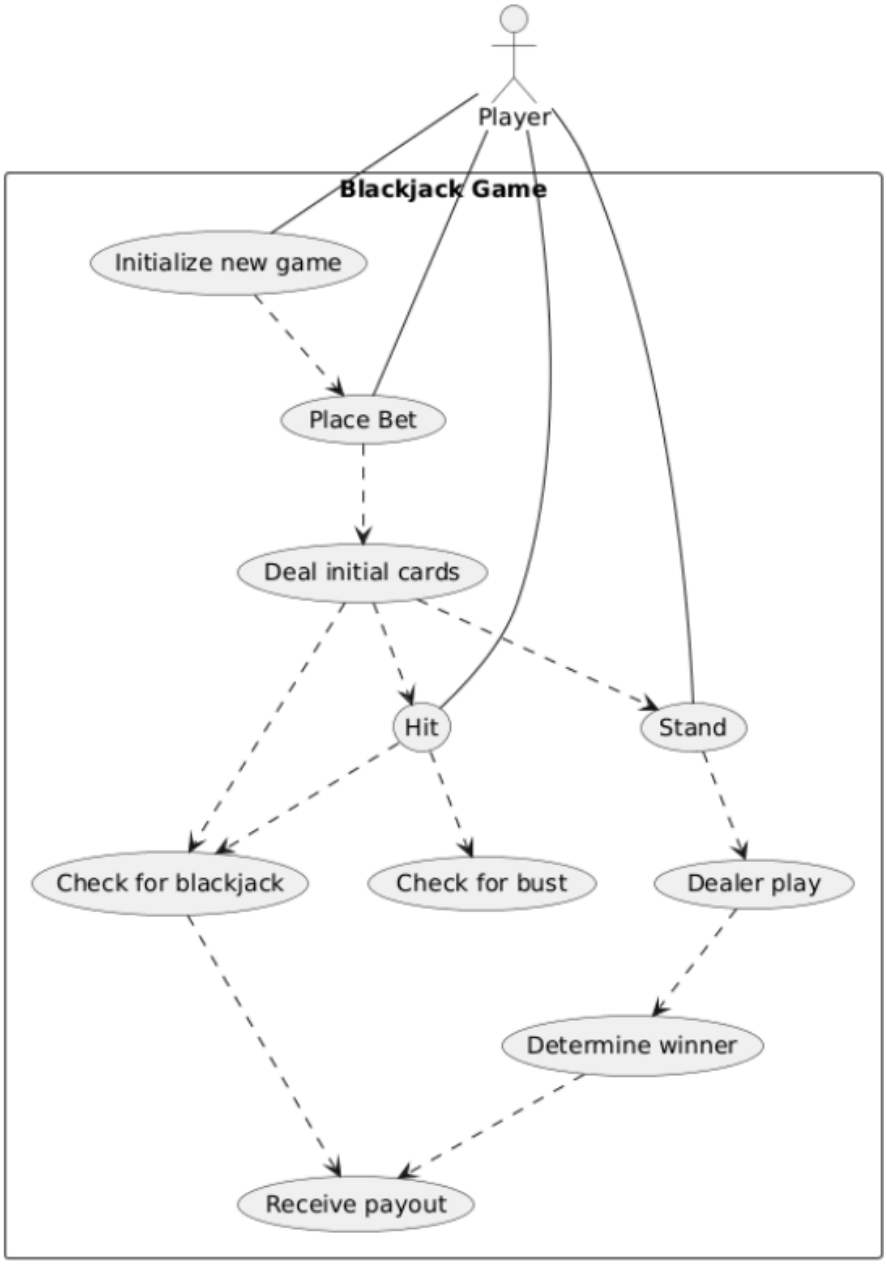
Blackjack is one of the most popular card games in the world and widely played in casinos. It is also extremely easy to learn the basics. The main goal of the game is to get as close to a hand with a value of 21 without going over. The player has very simple possible moves too. They can take another card or decline to do so and hope their hand is the closest one to 21. However, it is possible to make it as complex as a person wants. First, there are many side backs one can make with different rules and rewards. It is also possible to analyze strategy for which hands to make certain bets on.

The main motivation for building a Blackjack game in JavaFX is to provide individuals a safe, closed space to play the game. The environment is completely under their control. They are also able to safely explore Blackjack first to prepare to play it in the casino. It's a no-stakes, fun environment to learn the basic rules, get practice, and develop a strategy before facing the dealer in real life.

User Stories

As a player, I want the game to have all the basic functionalities of a real Blackjack game like placing a bet, drawing additional cards, doubling down on my bet, and tracking my winnings or losses.

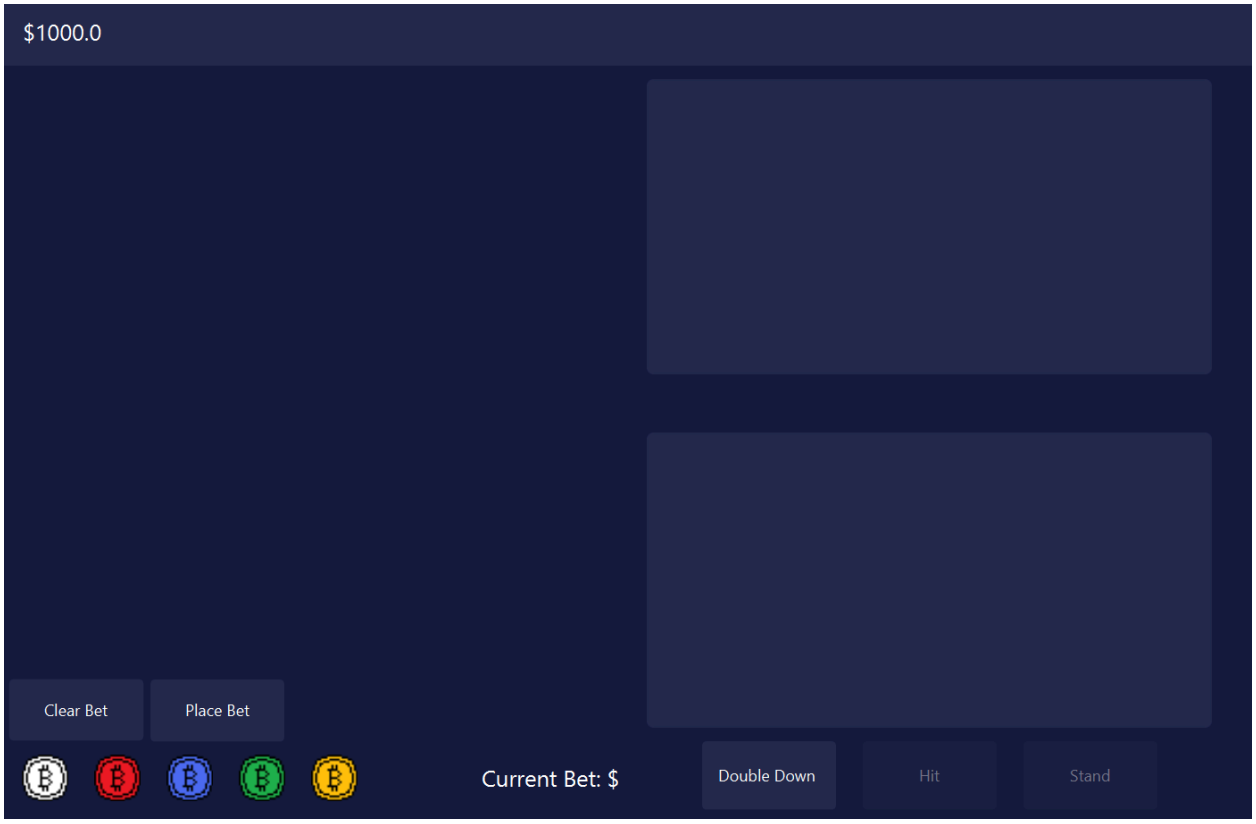
This use case diagram shows all the actions that the player can take as the game progresses.



Instruction

Start screen

- Blank space
- Currency set at $1,000



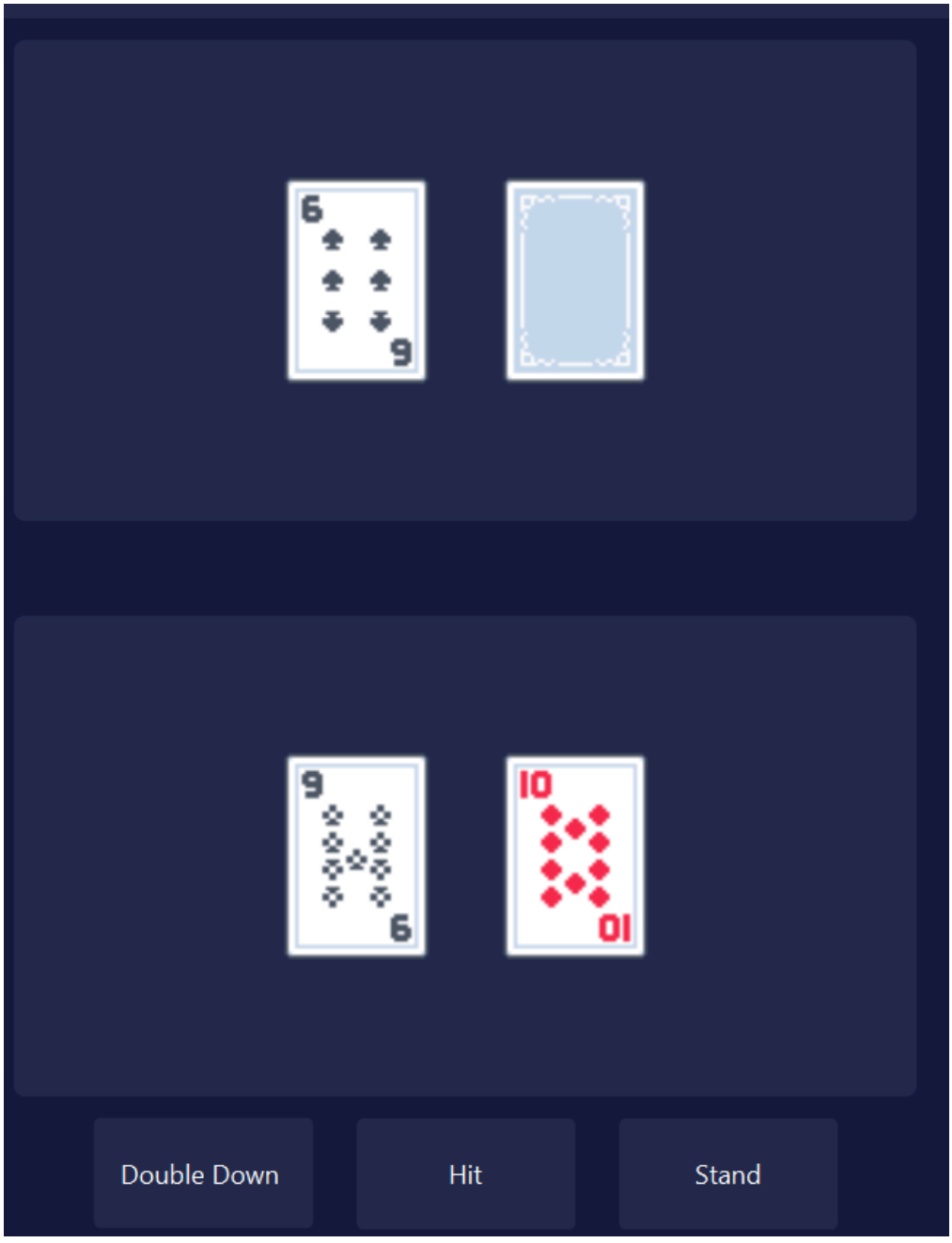
Placing bet

- Select currency
- Click “Place Bet”



Game screen

- Cards are shown
- Player can decide to hit, stand, or double down



Game ends

- Face losses or collect winnings
- Click “OK” to restart

